

10.) Injury to the sacro-iliac joint or the lumbo-sacral joint can lead to severe hindleg lameness. These areas are exposed to tremendous stress while racing, as all the force arising from the hind legs is translated into cranial movement of the body through these structures. Examination is performed by extending both hind legs simultaneously and applying pressure to the junction between the lumbar spine and the sacrum. Healthy animals will not react uncomfortably. Dogs with lumbo-sacral disease frequently react by sitting down suddenly when pressure is applied to the caudal back area. Severely affected animals may also experience pain upon lifting the tail, or show decreased anal sphincter tone.

11.) If the above listed examinations do not reveal a "sore spot", the cause of lameness may be secondary to a muscle injury, or secondary to pain from primary bone disease. While the former is common in racing sled dogs, the latter is not to be expected on the trail, but could be the cause for lameness in very young, severely injured, or in older, retired sled dogs. To localize the affected muscle, it is necessary to know its anatomical origin, course, and insertion. Closely associated with this knowledge is the nomenclature of bones and bony prominences of the canine skeleton. The most commonly injured muscles of the front leg are the biceps, triceps and extensor muscles. In the rear leg, common muscle injuries are the extensors, flexors, and rotators of the hip joint, as well as the gastrocnemius muscle. Deep, careful palpation of each of these muscles is necessary to localize the injured region. Chronic injuries (3-4 weeks) may lead to muscle atrophy on the affected leg. This may be detected when comparing the right and left leg. Chronic front leg lameness frequently results in atrophy of the shoulder muscles, which may be detected by a more pronounced acromion on the affected side. Primary bone diseases, such as panosteitis (inflammation of the bone) in young dogs, osteomyelitis (infection of the bone), for example after an open injury or surgery, or bone cancer (mainly in older animals) are frequently associated with a pain response upon deep palpation of the affected area of the long bones. Palpation of the long bones should be reserved for the last part of the orthopedic examination as bone pain can be extreme at times. The best locations to palpate the long bones without interfering with overlying musculature are the distal radius, proximal ulna, distal and proximal humerus, medial aspect of the femur, medial and lateral aspect of the stifle, and medial aspect of the tibia. Increasing pressure is applied to these areas and the animal is carefully observed for a reaction. In most cases with a positive response, radiographs should be taken to obtain a definitive diagnosis.

12.) Injuries to the spinal cord or peripheral nerves can also be the cause for lameness in sled dogs. The description of a complete neurologic examination is beyond the scope of this chapter, but information on this examination is widely available in the literature.⁶ A few basic steps include acute neurologic dysfunction, which may arise secondary to neck injuries which are seen mainly in teams using neck lines or after a tangle with dogs in the gangline; dogs may also get a leg caught in a loop of the gangline resulting in nerve damage or severe orthopedic damage. The composition of the harness may also play a role in the etiology of both neurological and orthopedic injuries of sled dogs. Animals with neck pain won't pull well, and will guard their neck when being examined. Besides ataxia (a wobbly gait), conscious proprioceptive deficits (the dog cannot feel where his feet are) are the earliest symptoms associated with neurological dysfunction. To evaluate conscious proprioception, the paw is knuckled over on its dorsal aspect, one foot at a time. Normal animals replace the foot immediately. If the flow of information from the paw to the brain is impaired, the paw will not be replaced to a normal stance immediately.

Some dogs may also injure the muscles surrounding the vertebral column or the vertebral bodies themselves. Depending on the degree of injury, affected dogs may show only pain upon palpation of the back, may not be able to coordinate all limbs, or may have difficulties moving one or several legs normally. To determine whether there is back pain, the examiner should palpate and apply pressure to the spinous processes of the vertebral bodies. The epaxial muscles on either side of the spinous processes should be palpated next. As there are no spinous processes on the cervical vertebral bodies, palpation of the neck is limited to deep palpation of the epaxial muscles, and the vertebral bodies. The vertebral bodies may be easier to locate in thin racing dogs, compared to regular pet dogs. Further examination of the neck includes careful flexion (nose down to the chest), extension (nose up into the air), and bending to the right and left (nose should touch the lateral chest wall on each side in a normal dog). It is critical to be very careful with these tests, as they may result in worsening of the condition when too much force is applied. Tumors of the nerves that supply the front leg may occur in older dogs and result in severe muscle atrophy of the front legs. Frequently a large, painful mass in the axillary area can be found upon deep palpation. Once more it is recommended to compare the right and left side of the standing dog to confirm the finding. Dogs with lumbo-sacral disorders may also show neurological abnormalities. The examination of this area is described above. Radiology, myelography, CT, or MRI may be necessary for a definitive diagnosis.

ADDITIONAL DIAGNOSTICS

It requires much practice to become an experienced examiner for orthopedic injuries. Using the techniques described above, it may not be possible to identify some injuries clearly. With unresolved, obscure lameness it may be worthwhile considering a consultation with an experienced veterinarian or veterinary surgeon who is interested in orthopedic injuries of the sled dog. Diagnostic imaging techniques, such as radiography, ultrasound, arthroscopy, CT, or MRI, as well as laboratory tests, such as blood work and/or joint fluid examination, may be helpful in definitively diagnosing the cause for lameness in the sled dog.

Table 1: Normal Joint Range of Motion in the Dog (Average of 10 mixed-breed dogs).⁷

Joint	Range of Motion (°)	Joint	Range of Motion (°)
Shoulder		Hip	
Flexion	60-70	Flexion	70-80
Extension	65-75	Extension	80-90
Hyperextension	0	Adduction	30-40
Adduction	40-50	Abduction	70-80
Abduction	40-50	Internal rotation	50-60
Internal rotation	40-50	External rotation	80-90
External rotation	40-50		
Elbow		Stifle	
Flexion	70-75	Flexion	65-75
Extension	70-75	Extension	65-75
Hyperextension	0	Hyperextension	0
Supination	80-90		
Pronation	40-50		
Carpus		Tarsus	
Flexion	155-160	Flexion	65-75
Extension	20-30	Extension	90-110
Ulnar deviation	15-20	Inversion	40-50
Radial deviation	15-20	Eversion	40-50

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